Common Error Messages in Python, NumPy, Pandas & Matplotlib

Handy one■page reference for typical exceptions and warnings you'll meet in day■to■day scientific Python work.

Core Python

SyntaxError

```
SyntaxError: invalid syntax
```

Typically caused by typos, missing punctuation, or mismatched parentheses/quotes.

IndentationError

```
IndentationError: unexpected indent
IndentationError: unindent does not match any outer indentation level
```

Occurs when indentation is inconsistent (mixing tabs/spaces or wrong block levels).

NameError

```
NameError: name 'x' is not defined
```

You referenced a variable or function before defining it, or it is misspelled.

TypeError

```
TypeError: unsupported operand type(s)
```

Operation applied to incompatible types (e.g., adding a string to an integer).

ValueError

```
ValueError: invalid literal for int()
```

Right type but bad value (e.g., int('abc')).

AttributeError

```
AttributeError: 'object' has no attribute 'x'
```

You tried to access an attribute or method that does not exist on the object.

IndexError

```
IndexError: list index out of range
```

List index exceeds current bounds.

KeyError

```
KeyError: 'some_key'
```

Dictionary key not found.

ImportError / ModuleNotFoundError

```
ImportError: cannot import name X
ModuleNotFoundError: No module named 'X'
```

Module/object missing, misspelled, or not on the import path.

ZeroDivisionError

```
ZeroDivisionError: division by zero
```

Attempted division by zero.

NumPy

Broadcasting error

```
ValueError: operands could not be broadcast together
```

Array shapes are incompatible for elementwise operations.

Ufunc type mismatch

```
TypeError: ufunc 'add' did not contain a loop with signature...
```

NumPy cannot operate on the provided dtypes (cast/convert as needed).

Indexing error

```
IndexError: too many indices for array
```

Indexing uses more dimensions than the array has.

Reshape error

```
ValueError: cannot reshape array of size X into shape Y
```

Element count does not match target shape.

Runtime warnings

```
RuntimeWarning: overflow encountered in ...
RuntimeWarning: divide by zero encountered in ...
```

Indicates numerical instability during operations.

Pandas

Column not found

```
KeyError: 'column_name'
```

Column label does not exist (typo, whitespace, or different columns after prior ops).

Too many indexers

```
IndexingError: Too many indexers
```

Tried to index a DataFrame/Series with excessive dimensionality.

Mismatched assignment length

```
ValueError: Length of values does not match length of index
```

Assigned array/list length differs from DataFrame index length.

SettingWithCopyWarning

```
A value is trying to be set on a copy of a slice...
```

Warning that a chained operation may modify a view, not the original object—use .loc and avoid chaining for assignment.

Concatenation type mismatch

```
TypeError: cannot concatenate object of type ...
```

Objects passed to concat are incompatible; ensure consistent types and shapes.

CSV parsing errors

```
ParserError: Error tokenizing data
```

Malformed CSV (delimiter/quotes/escape issues); set correct params like sep, engine, quoting.

Matplotlib (pyplot)

Mismatched x/y lengths

```
ValueError: x and y must have same first dimension
```

x and y arrays must have equal length for plotting.

Non■numeric data passed

```
TypeError: float() argument must be a string or a number
```

Plotting received non■numeric data—clean/convert inputs.

Wrong attribute access

```
AttributeError: 'AxesSubplot' object has no attribute ...
```

Mixing pyplot stateful API with OO API incorrectly; use returned Axes methods consistently.

No GUI backend available

```
RuntimeError: Invalid DISPLAY variable
```

Headless environment without GUI backend; switch backend (e.g., 'Agg') or use inline in notebooks.

Missing font glyph

```
UserWarning: Glyph X missing from current font
```

Current font lacks character; install/choose a font that supports it or use rcParams.

Tip: Keep this sheet nearby when debugging; search the exact message for context■specific fixes.